KhetiConnect: Direct Farmer-to-Consumer Marketplace, Education and Smart Logistics

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Abstract—KhetiConnect is a digital platform designed to improve direct market access for farmers. It connects farmers with buyers, supports free produce listing for farmers, and provides fixed-price and auction-based selling. The platform also offers agricultural education, government-scheme information, market-price information, and order/transport tracking. Its proposed revenue model charges buyers/consumers and related services rather than farmers. The prototype is being developed with React, Vite, Tailwind CSS, Express.js, a local JSON data store, JWT authentication, and bcrypt password hashing.

Index Terms— farmer marketplace, direct trade, agricultural education, market information, transport tracking, crop bidding.

1 Introduction

Agriculture continues to be one of the largest sources of rural livelihood in India, yet the path between a farmer's field and a buyer's table is rarely direct. Produce typically changes hands through village-level aggregators, commission agents, and wholesale traders before reaching a consumer, and each intermediary stage adds cost while reducing the price ultimately received by the farmer. At the same time, buyers often have little visibility into where their produce comes from, how it was grown, or whether the price they pay is fair.

Beyond market access, farmers frequently lack convenient access to three other resources that materially affect their income: up-to-date mandi (market) price information, structured agricultural knowledge and advisory content, and dependable visibility into transport once produce leaves the farm. Existing digital solutions tend to solve only one of these problems at a time — a trading portal, a price-information app, or an advisory service — which forces farmers to move between multiple, unconnected tools.

KhetiConnect is proposed as a single, farmer-first web platform that unifies direct produce trading, agricultural education, government-scheme awareness, live market-price information, and order/transport tracking. The guiding design principle is that farmers should never be charged to access the platform or to list their produce; instead, the platform's proposed revenue is drawn from buyers and from optional value-added services.

The remainder of this paper is organized as follows. Section 2 discusses the background and motivation behind the project. Section 3 surveys related work and situates KhetiConnect against existing agri-tech platforms. Section 4 describes the system architecture and objectives. Section 5 explains the implementation methodology and module design. Section 6 discusses the development timeline, current results, and testing approach. Section 7 outlines limitations and future scope, followed by the conclusion.

2 Background and Motivation

2.1 Key challenges

- **Limited market access** – dependence on local intermediaries and a small, fixed set of known buyers reduces a farmer's ability to discover better prices or reach buyers outside their immediate area.
- **Information gaps** – convenient, farmer-friendly access to current mandi prices, weather-appropriate farming practices, and applicable government schemes is often scattered across notice boards, phone calls, and unrelated apps.
- **Logistics visibility** – once produce is dispatched, farmers and buyers frequently have no shared way to see where an order is in its journey, which creates uncertainty on both sides of a transaction.
- **Fragmented digital services** – trading, learning, and market-information tools exist independently of one another, so a farmer who wants all three must learn and maintain accounts on several different platforms.
- **Trust deficit** – buyers cannot always verify who a farmer is or how reliable they have been in the past, and farmers face the same uncertainty about unfamiliar buyers, which discourages first-time direct transactions.

2.2 Motivation and scope

The motivation for KhetiConnect is to reduce the number of separate systems a farmer must depend on by offering selling, learning, price-checking, and order-tracking within one coherent interface. The project is scoped as an academic web-application prototype: it demonstrates the full user experience and core business logic for all major workflows, while treating production-grade concerns such as live external data feeds, real payment settlement, and third-party logistics integration as future work rather than part of the current deliverable.

This scoping decision keeps the prototype achievable within a single academic semester while still allowing every core actor — farmer, buyer, and administrator — to complete an end-to-end journey through the system.

3 Literature Survey and Comparative Analysis

A structured review of existing digital agriculture initiatives was carried out to understand which marketplace and farmer-support features already exist, and where gaps remain that KhetiConnect can address.

3.1 Farmer-Focused Digital Services (Miraitu)

Miraitu [3] was reviewed as a reference point for farmer-oriented digital services built around agronomic support and farmer outreach. Platforms of this kind typically focus on providing farmers with informational and advisory support rather than an integrated, buyer-facing produce-trading experience, which limits their ability to directly close the gap between production and sale.

3.2 Farm-Input and Advisory Platforms (BigHaat)

BigHaat [2] was reviewed as a reference point for farmer-oriented digital services, offering farm inputs, crop-advisory content, and agri-commerce features. Such platforms strengthen a farmer's access to inputs and guidance but are generally organized around supplying the farmer rather than around selling the farmer's own produce directly to a buyer.

3.3 Online Agricultural Marketplaces (Kisan Mandi)

Kisan Mandi [1] represents a category of online platforms built around farm-product discovery and trade. Such platforms demonstrate that a searchable, browsable produce catalogue is a workable model for digital agricultural commerce, but they generally treat trading as a standalone function, without embedded advisory content, mandi-price context, or transport visibility.

3.4 Transportation and Market-Channel Research

Independent research on agricultural market-channel choice in India has identified transportation and communication infrastructure as significant factors influencing which markets a farmer can realistically access and the price they are able to negotiate. This supports the design decision to make order and transport tracking a first-class feature of KhetiConnect rather than an afterthought.

3.5 Comparative Positioning

Table 1 summarizes how KhetiConnect is positioned relative to the platforms reviewed above. The consistent theme across the literature is that existing systems each address one dimension of the problem well — market access, advisory content, or product discovery — but none combine direct trade, education, market-price awareness, and logistics visibility inside a single, farmer-first ecosystem. This is the specific gap that KhetiConnect is designed to close.

4 System Architecture and Objectives

4.1 Core objectives

The primary objective is to design and build a web platform that connects farmers directly with buyers without charging farmers to list or sell their produce. This primary objective is supported by a set of secondary objectives:

- Provide both fixed-price and live-auction (bidding) listing formats so farmers can choose the selling method that suits their produce and urgency.
- Deliver an educational and advisory hub containing farming guides, instructional videos, and structured content on relevant government schemes.
- Surface market/mandi price information so farmers can make more informed pricing decisions before listing produce.
- Provide order and transport tracking so both farmers and buyers can see the status of a transaction after it is placed.
- Offer role-specific dashboards for farmers, buyers, and administrators, and build trust through verified profiles, ratings, and reviews.

4.2 System actors and platform workflow

The system defines four actors: Farmer, Buyer, Administrator, and Transport Partner. A farmer registers, builds a profile, and creates produce listings in either a fixed-price or auction format. A buyer browses or searches the marketplace, views product and farmer details, and either places a direct order or submits a bid on an auction listing. Once an order is confirmed, both parties can follow its progress through defined stages until delivery, after which the buyer may rate and review the farmer. Administrators oversee user verification, content moderation for the advisory hub, and overall platform activity.

4.3 Revenue model

Farmers are not charged for registration, produce listing, or use of the advisory and market-information features. The proposed revenue model instead derives income from buyers and consumers, through mechanisms such as a service fee on completed transactions or optional delivery-related charges, so that the cost of participating in the platform does not fall on the producer.

4.4 UML system modeling

The system was modeled using standard UML artifacts before implementation began. The use-case diagram captures the distinct interactions available to Farmers, Buyers, and Administrators. The class diagram defines the core entities — User, Farmer, Buyer, Product, Category, Order, Payment, Delivery, and Transport Partner — along with their attributes and relationships. A sequence diagram traces the order-placement workflow from product selection through payment and delivery confirmation, and an activity diagram maps the decision points in the fixed-price versus auction selling flow. Together these diagrams guided the database schema and API design used in the implementation.

5 Module Implementation and Methodology

5.1 Technology stack

The prototype follows a component-based web architecture. The frontend is built with React 18 and Vite for fast development and hot module reloading, styled using Tailwind CSS utility classes, with Lucide React supplying iconography and Canvas Confetti used for lightweight success-state animations (for example, on order confirmation). The backend is implemented with Express.js and currently persists data in a local JSON-based data store, chosen for its simplicity during prototype development; migrating this store to a proper relational or document database is planned as the next implementation step. Authentication is handled using JSON Web Tokens (JWT), and user passwords are hashed using bcrypt before storage so that plaintext credentials are never persisted.

5.2 Module breakdown

The implementation is organized into distinct functional modules, each corresponding to a defined area of the use-case model:

- **Authentication and role management** – registration, login, and role-based access control distinguishing Farmer, Buyer, and Admin sessions.
- **Marketplace and product listing** – creation, browsing, search, and filtering of produce listings with category and location context.
- **Fixed-price and auction workflow** – direct-order checkout for fixed-price listings, and a bidding interface with bid history and farmer acceptance for auction listings.
- **Advisory hub** – structured farming guides, instructional videos, and government-scheme information, organized by topic.
- **Market-price section** – a reference view of mandi/market price information to support farmer pricing decisions.

- **Dashboards** – role-specific views summarizing a farmer's active listings and orders, or a buyer's order history and tracked deliveries.
- **Order tracking and reviews** – order-status progression from placement to delivery, and a post-delivery rating and review mechanism.

5.3 Data design

The data model reflects the class diagram produced during system modeling. A User record carries shared account information and a role flag; Farmer and Buyer extend this with role-specific fields such as farm location or delivery address. A Product belongs to a Farmer and references a Category, and carries a selling-type flag distinguishing fixed-price from auction listings. An Order links a Buyer, a Product, and a Payment record, and holds a Delivery sub-record that is updated as a Transport Partner progresses the order through its stages.

5.4 Security considerations

Two security measures are already implemented in the prototype: bcrypt-based password hashing, which ensures that stored credentials cannot be trivially reversed even if the data store is exposed, and JWT-based session authentication, which allows API routes to verify a user's identity and role on every request without storing session state on the server. Input validation on registration and listing forms is applied at the frontend; adding matching server-side validation and rate-limiting on authentication endpoints is identified as near-term hardening work before any move beyond prototype deployment.

6 Timeline, Results and Discussion

6.1 Development timeline

The project follows a semester-long plan spanning August to December. Problem identification and literature survey were carried out in August and September; requirement analysis and system design (including the UML modeling described in Section 4.4) ran through September and October; frontend development proceeded in parallel from September into October; backend and data-store integration is scheduled for October and November; testing is planned for November and December; and final documentation and presentation close out the semester in December.

6.2 Current results

At the present stage, the prototype demonstrates a working marketplace interface, complete authentication and role-based access, fixed-price and auction listing creation and browsing, and the advisory-hub and market-price sections. Backend integration with the JSON data store and the transport-tracking module are in progress at the time of writing. Production-readiness items — a persistent production database, a real payment gateway, live external mandi-price data, and third-party logistics APIs — remain outside the current implementation and are treated as future work rather than claimed results.

6.3 Testing approach

Verification to date has consisted of manual functional testing of each module in isolation — confirming that registration, login, listing creation, bidding, order placement, and dashboard views behave correctly for each of the three user roles — together with informal usability walkthroughs with a small number of prospective farmer and buyer users to surface confusing or unclear interface elements. Formal, structured test cases and quantitative usability metrics are planned for the testing phase of the timeline described above; no performance or adoption figures are reported at this stage, since none have yet been measured.

6.4 Discussion

The results so far indicate that the core interaction model — a single farmer-first platform spanning trade, education, market information, and logistics — is implementable within a conventional React/Express web stack without requiring specialized infrastructure. The main open engineering question is the migration from the current local JSON data store to a database that can support concurrent writes safely once multiple farmers and buyers use the platform simultaneously, which is the immediate priority for the next development phase.

Table 1: COMPARATIVE ANALYSIS OF AGRITECH PLATFORMS

Platform	Focus	KhetiConnect Contribution
Kisan Mandi [1]	Online farm-product discovery and trade	Adds advisory and transport tracking
BigHaat [2]	Farm-inputs and advisory-focused digital services	Combines trade, education and tracking
Miraitu [3]	Farmer-oriented advisory and outreach services	Adds direct trade and logistics visibility
KhetiConnect	Integrated marketplace (web prototype)	Direct trade + education + market info + logistics

Table 2: MODULE IMPLEMENTATION STATUS

Module	Technology / Method	Status
Authentication & Roles	React, JWT, bcrypt	Completed
Marketplace & Listing	React, Tailwind CSS	Completed
Fixed-price & Bidding	React state / custom logic	Completed
Advisory Hub	React components	Completed
Orders & Dashboard	Express, JSON store	In Progress
Transport Tracking	Prototype workflow	In Progress
Testing & Deployment	Manual / local	Pending

Work Carried Out Till Date

Area	Work Carried Out
Conceptual	Problem and solution identified
Analytical	Requirements and features analyzed
Theoretical	Existing platforms studied
Computational	UI and website modules developed
Methodological	Farmer and buyer workflows planned
Boundary	Prototype-based implementation

The implementation boundary is limited to the academic prototype; production deployment, external payment gateways, live market APIs, and commercial logistics integrations are outside the current scope.

7 Limitations and Future Scope

7.1 Current limitations

As an academic prototype, KhetiConnect has several limitations that are important to state explicitly rather than gloss over. The data store is a local JSON file rather than a production database, which does not scale to concurrent multi-user access. Payment is represented by a prototype checkout flow rather than an integrated payment gateway, so no real financial transactions occur. Market-price information is illustrative rather than sourced from a live mandi-price feed, and transport tracking currently follows a simulated workflow rather than data from an actual logistics partner. Formal security testing, load testing, and accessibility auditing have not yet been performed.

7.2 Future scope

- **Multilingual support** – Gujarati, Hindi, and other regional-language interfaces to widen accessibility for farmers.
- **Verified farmer onboarding** – document-based identity and land-record verification to increase buyer trust.
- **Live mandi-price integration** – replacing illustrative price data with a live feed from a government or market-data API.
- **Payment-gateway integration** – replacing the prototype checkout with a real payment gateway supporting UPI, cards, and net banking.
- **Logistics API integration** – connecting with real transport partners for GPS-based, real-time delivery tracking.
- **Database migration** – moving from the local JSON store to a production-grade relational or document database.
- **Advanced advisory features** – AI-assisted crop guidance, localized weather alerts, and voice-based interaction for lower-literacy users.

8 Conclusion

KhetiConnect presents a farmer-first digital platform for direct farmer-to-buyer access that combines a marketplace with farmer education, government-scheme information, market-price awareness, and order/transport tracking within a single interface. By allowing farmers to list and sell produce without any platform or listing charge, and by deriving proposed revenue from buyers and related services instead, the design keeps the platform accessible to the smallholder farmers it is intended to serve.

The current prototype, built with React, Vite, Tailwind CSS, and Express.js, demonstrates the complete user experience for farmers, buyers, and administrators across authentication, listing, bidding, ordering, advisory content, and dashboards. Backend persistence and transport tracking are the immediate next steps, and the future-scope items identified in Section 7.2 — particularly live market data, real payment integration, and verified onboarding — represent the path from academic prototype toward a deployable system capable of making a measurable difference in farmers' direct market access.

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References

[1] Kisan Mandi, https://www.kisanmandi.com/

[2] BigHaat, https://www.bighaat.com/

[3] Miraitu, https://www.miraitu.in/

About the Authors

Kushwaha Ankitkumar is a B.Tech student in Computer Science & Engineering at Silver Oak University, who conceived the original KhetiConnect concept and directed its overall design and development. He identified the core problem of fragmented farmer support — scattered trading, advisory, and logistics tools — and shaped the platform's farmer-first approach, including the no-cost listing model and the combination of marketplace, education, and tracking features into a single system.

Jeet Ranpariya is a B.Tech student in Computer Science & Engineering at Silver Oak University, who developed the KhetiConnect website, building the React and Express-based prototype, including the marketplace interface, authentication flow, and the fixed-price and auction listing workflows described in this paper.

Het Nakrani is a B.Tech student in Computer Science & Engineering at Silver Oak University, who worked on system modeling, contributing to the use-case, class, and sequence diagrams that define the platform's actors, entities, and core workflows.

Yash Makwana is a B.Tech student in Computer Science & Engineering at Silver Oak University, who worked on application development, contributing to the implementation of the advisory hub, dashboard views, and supporting frontend components.

Rajdipsinh Chauhan is a B.Tech student in Computer Science & Engineering at Silver Oak University, who worked on the user interface and supporting project modules, contributing to the visual design and usability of the marketplace and dashboard screens.